

about surveillance and portable processing needs for weapon-mounted sights, handheld range and target finders, and unmanned air or ground platforms, the module should first meet the demanding size, weight and power constraints. The combination of dedicated image-processing cores and an abundance of peripherals in a single integrated circuit has resulted in all-in-one image processors with a general-purpose computer. These can be useful for performing other critical non-vision tasks, such as flight control and system-to-system communication.

The optimised real-time vision-processing hardware performs operations such as noise reduction, image enhancement, image fusion, stabilisation and object tracking.

HDMI transceivers

These transceivers have evolved offering flexibility for your designs. These allow you to separate the audio from video or *vice versa*. Such transceivers

let the video pass through and bring out a compressed audio on the device for DSP to process. A good thing about such devices is that these are not only useful for home audio and video but also for professional AVs.

Subramanyam of Analog Devices, India says, "You can take the audio out from these transceivers and reinsert it back, which is a kind of requirement for studio equipment. It offers a kind of great flexibility, taking audio from one channel and putting audio and video on different locations."

Smart amplifiers

An amplifier is an integral part of an audio system. Digital amplifier products, where analogue quality performance, reduced size and improved energy efficiency are at premium, have been the focus of the market in recent times. As a result, the 'smart' amplifiers have surfaced for helping the design engineers. These smart amplifiers amplify the signal with constant

monitoring of the functional device, thereby protecting it from damage and at the same time ensuring the quality of the output.

Ganapathy explains, "The smart amplifier technology allows you to have a 5W speaker with 10x the range without damaging the speakers, so with 5W you can have the 50W experience and thus have more compact designs. It continuously monitors speaker characteristics and responds accordingly; this is how you get good audio tones without damaging the speakers."

Over the years use of smart devices has become common, but the potential interference they generate has always retarded the aim of providing an excellent sound performance. The new amplifiers that have come up for the smart devices, provide increased GSM robustness to mitigate the influence of smartphones on loudspeakers, resulting in a cost-effective solution that improves sound quality.



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